13. A Model for SQA Defect Removal Effectiveness and Cost

13.1 Introduction to a model for SQA

13.2 Calculation of Standard and Comprehensive SQA plan

13.3 Calculation of Economic SQA plan

1. What is a model for SQA?

The model deals with two quantitative aspects of an **SQA plan** consisting of **several defect detection activities**:

1. The plan's total effectiveness in removing project defects
2. The total costs of removal of project defect

Note

SQA activities must be integrated within the project's development plan.

2. The Data

The application of the model is based on **three types of data**:

1. **Defect origin distribution**
2. **Defect removal effectiveness**
3. **Cost of defect removal**

1) Defect Origin Distribution

- Defect origins are distributed throughout the development process
- Software development professionals reveal a similar pattern of defect distribution.
- They believe that this pattern has not changed substantially in the last two decades.

A Characteristic Distribution of Software Defect Origins

No	Software Development Phase	Average percentage of defects originating Phase
1	Requirement specification	15%
2	Design	35%
3	Coding (coding 30%, integration 10%)	40%
4	Documentation	[redacted] 10%

2) Defect Removal Effectiveness

- It is assumed that any quality assurance activity filters a certain percentage of existing defects.
- In most cases, the percentage of removed defects is somewhat lower than the percentage of detected defects
- The remaining defects, those undetected and uncorrected, are passed to successive development phases.

2) Defect Removal Effectiveness (Cont'd)

- The next quality assurance activity applied confronts a combination of defects

- It is assumed that the filtering effectiveness of accumulated defects of each quality assurance activity is not less than 40%.

2) Defect Removal Effectiveness (Cont'd)

- Removal efficiency is the most important quality measure

$$\text{Removal efficiency} = \frac{\text{Defects found}}{\text{Defects present}}$$

"Defects present" is the critical parameter

2) Defect Removal Effectiveness (Cont'd)

1	2	3	4	5	6	7	8	9	10

Defects

First operation 6 defects from 10 or 60% efficiency

Second operation 2 defects from 4 or 50% efficiency

Cumulative efficiency 8 defects from 10 or 80% efficiency

2) Defect Removal Effectiveness (Cont'd)

Defect Removal Efficiency = Percentage of defects removed by a single level of review, inspection or test

Cumulative Defect Removal Efficiency = Percentage of defects removed by a series of review, inspection or test

2) Defect Removal Effectiveness (Cont'd)

DEVELOPMENT DEFECTS		
Inspections	500	
Testing		400
Subtotal		900
USER-REPORTED DEFECTS IN FIRST 90 DAYS		
Valid unique		100
TOTAL DEFECT VOLUME		
Defect totals		1000
REMOVAL EFFICIENCY		
Dev. (900) / Total (1000) = 90%		

2) Defect Removal Effectiveness (Cont'd)

No	Quality assurance activity	Average defect Filtering effectiveness rate
1	Requirements specification review	50%
2	Design inspection	60%
3	Design review	50%
4	Code inspection	65%
5	Unit test	50%
6	Unit test after code inspection	30%
7	Integration test	50%
8	System tests/acceptance test	50%
9	Documentation review	50%

3) Cost of Defect Removal

- Removal of defects differs very significantly by development phase.

- Cost are much greater in later development phases.

  For example, the design phase may require 2.5 working days; during the acceptance tests require 40 working days

3) Cost of Defect Removal (Cont'd)

- In general, defect removal data is not commonly available.

- Most agree with the data based on key studies.

Representative Average Relative Defect-Removal Costs

No	Software Development Phase	Average percentage of defects Cost
1	Requirement specification	1
2	Design	2.5
3	Unit tests	6.5
4	Integration tests	16
5	System tests /acceptance tests/System documenta-tion review	40
6	Operation by Customer	110

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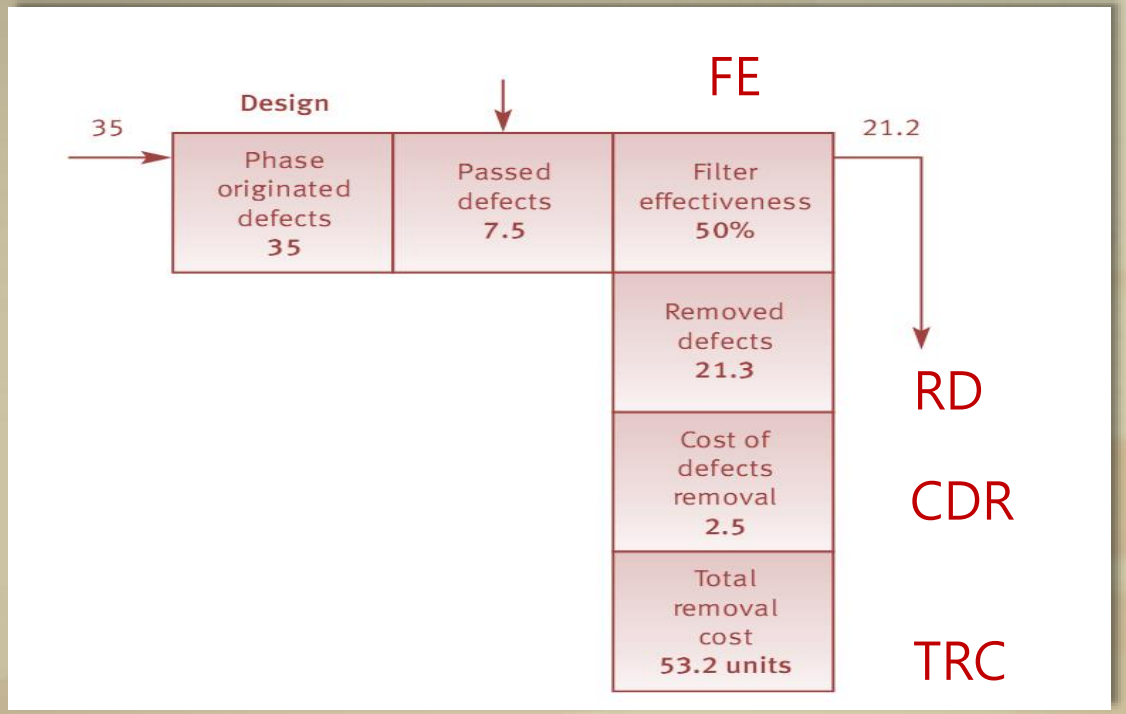
Model Assumptions

1. The development process is **linear** and **sequential.**
2. A number of **“new” defects** are introduced in each development phase.
3. **Removing a percentage** of the entering defects and letting the rest pass to the next development phase.
4. At each phase, **incoming defects** are the **sum** of those not removed **plus** new defects in current phase.
5. The **cost of defect removal** is calculated for each QA activity by **multiplying** the number of defects removed by the relative cost
6. The **remaining defects** passed to the customer, their full removal at this phase will **incur heavy costs.**

2. Model Parameters

POD	Phase Originated Defects
PD	Passed Defects (from former phases)
%FE	% of Filtering Effectiveness
RD	Removed Defects
CDR	Cost of Defect Removal
TRC	Total Removal Costs

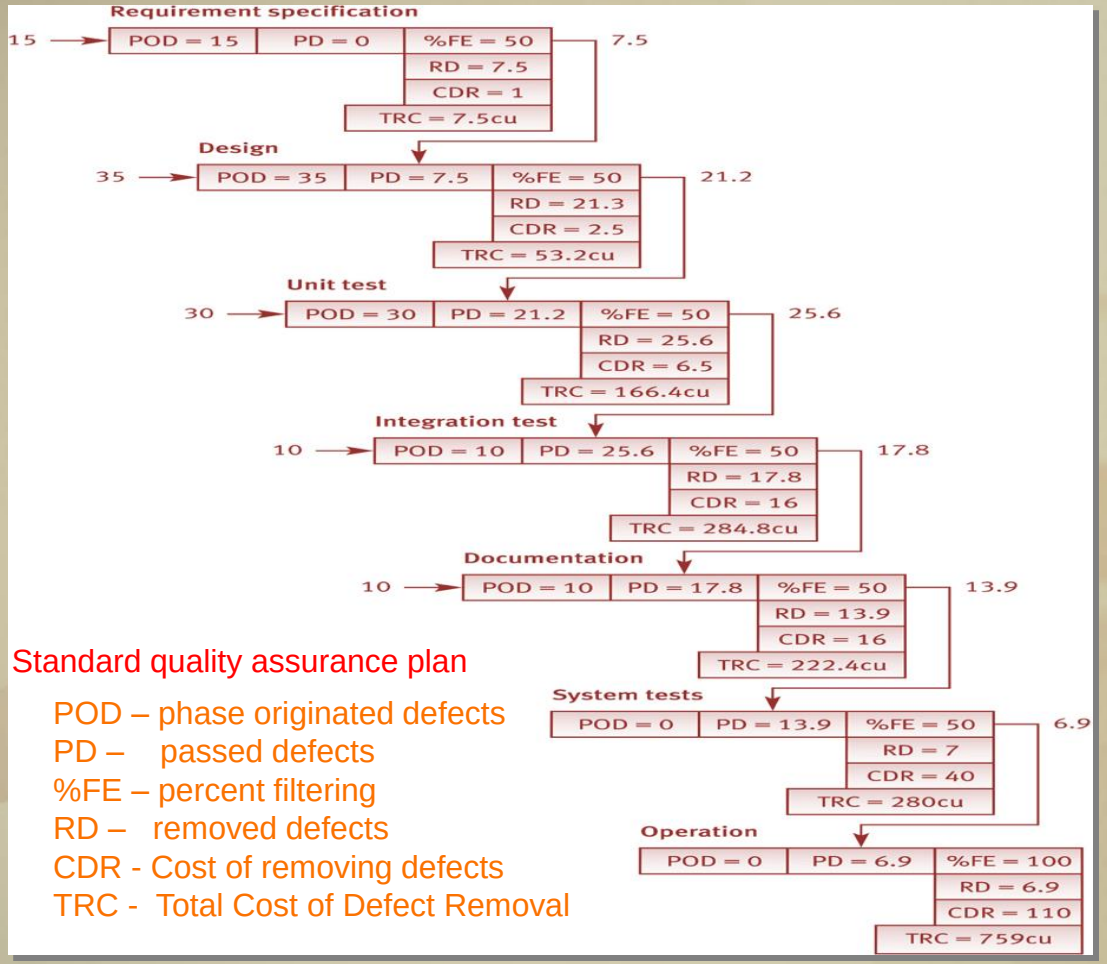


TRC = Total Removal Cost: TRC = RD × CDR

4. First Model (Standard Plan)

No	Quality assurance activity	Defect removal effectiveness	Cost of removing a detected defect
1	Requirement specification review	50%	1
2	Design review	50%	2.5
3	Unit test – code	50%	6.5
4	Integration test	50%	16
5	Documentation review	50%	16
6	System test	50%	40
7	Operation phase	100%	110

4. First Model (Standard Plan) (Cont'd)



Standard quality assurance plan

POD – phase originated defects
PD – passed defects
%FE – percent filtering
RD – removed defects
CDR - Cost of removing defects
TRC - Total Cost of Defect Removal

5. Comprehensive QA Plan

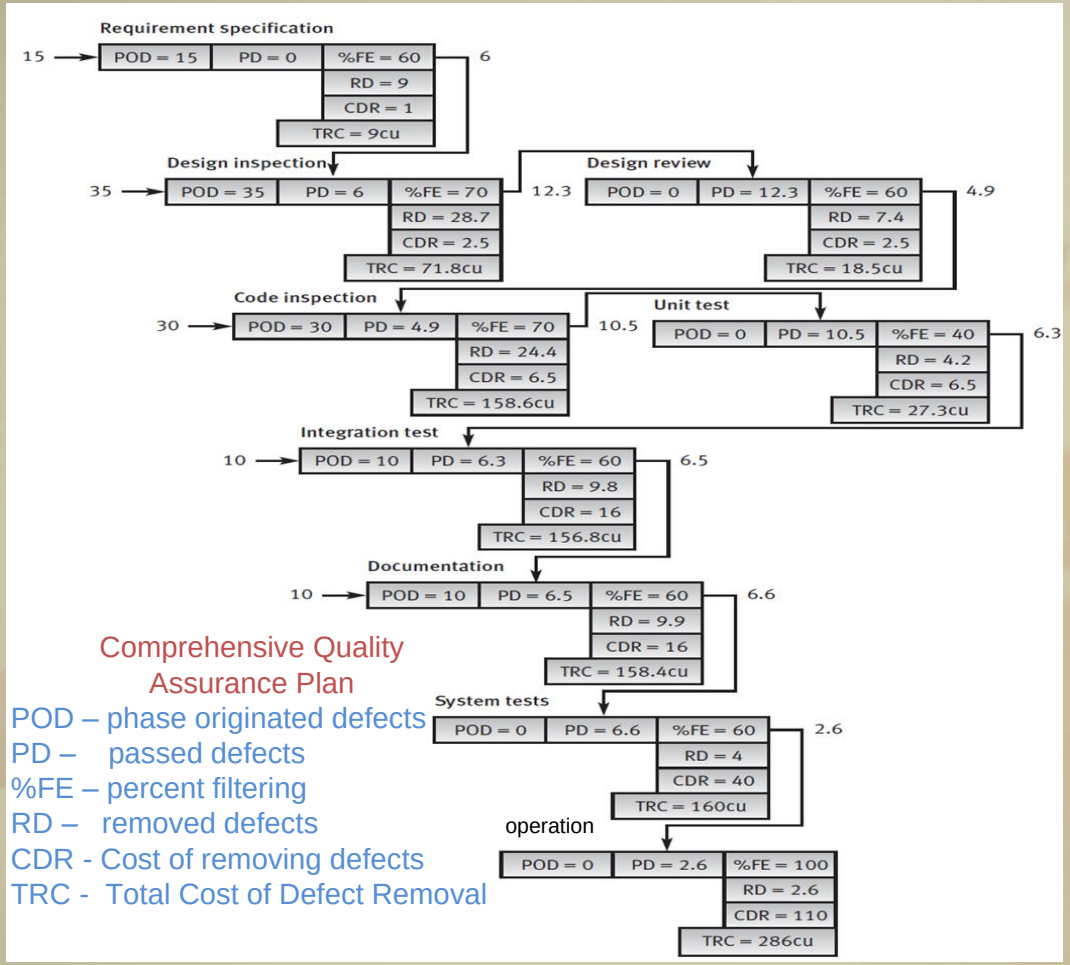
The comprehensive quality assurance plan (comprehensive defects filtering system) accomplishes the following:

1) Adds two quality assurance activities so that the two are performed in the design phase as well as in the coding phase
2) Improves the ‘filtering’ effectiveness of other quality assurance activities.

5. Comprehensive QA Plan (Cont'd)

No	Quality assurance activity	Defect removal effectiveness	Cost of removing a detected defect (cost units)
1	Requirement specification review	60%	1
2	Design inspection	70%	2.5
3	Design review	60%	2.5
4	Code inspection	70%	6.5
5	Unit test – code	40%	6.5
6	Integration test	60%	16
7	Documentation review	60%	16
8	System test	60%	40
9	Operation phase	100%	110

5. Comprehensive QA Plan (Cont'd)



Comprehensive Quality Assurance Plan

POD – phase originated defects
PD – passed defects
%FE – percent filtering
RD – removed defects
CDR - Cost of removing defects
TRC - Total Cost of Defect Removal

6. Comparison of the standard and Comprehensive quality assurance plans

No	Quality Assurance Activity	Standard Plan		Comprehensive Plan	
		Percentage of removed defects	Cost of removing defects	Percentage of removed defects	Cost of removing defects
1	Requirement Specification review	7.5%	7.5	9%	9
2	Design inspection	-	-	28.7%	71.8
3	Design review	21.3%	53.2	7.4%	18.5
4	Code inspection	-	-	24.4%	156.8
5	Unit test – code	25.6%	166.4	4.2%	27.3
6	Integration test	17.8%	284.8	9.8%	156.8
7	Documentation review	13.9%	222.4	9.9%	158.4
8	System test	7.0%	280	4%	160
	Total for internal quality assurance activities	**93.1%**	**1014.3**	**97.4%**	**760.4**
	Defects detected during operation	6.9%	759	2.6%	286
	Total	100.0%	**1773.3**	100.0%	**1046.4**

Standard plan

- removes 57.6% of the defects in requirements and design

Comprehensive plan

- removes 90.2% of the defects in requirements and design
- much more economical
- saves 41% of the total resources investing in defect removal
- a greater contribution to customer satisfaction by drastically reducing the rate of defects detected

Additional investments in QA activities yield **substantial savings in defect removal costs**

8. Utilization of the model

The model allows **calculating estimates of the cost** of decisions regarding the quality assurance plan

- Addition or elimination of a quality assurance activity from a given plan.
- Application of current quality assurance procedures activity versus application of a more efficient yet more costly procedure.

Utilization of the model enables **comparison of SQA policies and activity plans**.

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1. An Economy Plan

Limits the **quality assurance activities** to a **standard design review**, as required by the contract with the customer (50% filter)

A comprehensive system tests (60% filter)

Operation phase detection (100%)

Software Development Phase	Average % of defects originating phase	Average relative defect removal cost
Requirement specification	15%	1
Design	35%	2.5
Unit testing	30%	6.5
Integration testing	10%	16
Documentation	10%	40
System testing	-----	40
Operation	-----	110

3. Defects Removal Effectiveness for Quality Assurance Plans

Quality assurance activity	Defects removal effectiveness for economy SQA plan	Defects removal effectiveness for standard SQA plan	Defects removal effectiveness for Comprehensive SQA plan
Requirement specification review	-----	50%	60%
Design inspection	-----	-----	70%
Design review	50%	50%	60%
Code inspection	-----	-----	70%
Unit test	-----	50%	40%
Integration tests	-----	50%	60%
Documentation review	-----	50%	60%
System test	60%	50%	60%
Operation phase detection	100%	100%	100%

3. Defects Removal Effectiveness for Quality Assurance Plans (Cont'd)

Defect removal phase	Defect removal effectiveness	Average relative defect removal cost {cost unit}				
		Defect origination phase				
		Req	Des	Uni	Int	Doc
Requirement specification (Req)	50%	1	---	---	---	---
Design (Des)	50%	2.5	1	---	---	---
Unit testing (Uni)	50%	6.5	2.6	1	---	---
Integration (Int) System documentation (Doc)	50% 50%	16 16	6.4 6.4	2.5 2.5	1	 1
System testing / Acceptance testing (Sys)	50%	40	16	6.2	2.5	1
Opertion by customer (after release)	100%	110	44	17	6.9	2.5

4. An Economy Quality Assurance Plan based on the SQA plan

Requirement specification (POD=15)

	Ddoc	Dint	Duni	Ddes	Dreq	Total		
ID					15	15		15
PD					15	15	%FE=0	
RD					0	0		
RDRC					1	TRC >	0 cu	

Design (POD=35)

	Ddoc	Dint	Duni	Ddes	Dreq	Total		
ID				35	15	50		25
PD				17.5	7.5	25	%FE=50	
RD				17.5	7.5	25		
RDRC				1	2.5	TRC >	36.3 cu.	

Unit tests (POD=30)

	Ddoc	Dint	Duni	Ddes	Dreq	Total		
ID			30	17.5	7.5	55		55
PD			30	17.5	7.5	55	%FE=0	
RD			0	0	0	0		
RDRC			1	2.6	6.5	TRC >	0 cu	

Integration tests (POD=10)

	Ddoc	Dint	Duni	Ddes	Dreq	Total		
ID		10	30	17.5	7.5	65		65
PD		10	30	17.5	7.5	65	%FE=0	
RD		0	0	0	0	0		
RDRC		1	2,5	6,4	1.6	TRC >	0 cu	

4. An Economy Quality Assurance Plan based on the SQA Plan (Cont'd)

Documentation reviews (POD=10)

	Ddoc	Dint	Duni	Ddes	Dreq	Total	
ID	10	10	30	17.5	7.5	75	
PD	10	10	30	17.5	7.5	75	%FE=0
RD	0	0	0	0	0	0	
RDRC	1	1	2.5	6.4	16	TRC >	0 cu

75

System tests (POD=0)

	Ddoc	Dint	Duni	Ddes	Dreq	Total	
ID	10	10	30	17.5	7.5	75	
PD	4	4	12	7	3	30	%FE=60
RD	6	6	18	10.5	4.5	45	
RDRC	2.5	2.5	6.2	1.6	40	TRC >	48.9.6 cu

30

Operation by customer (POD=0)

	Ddoc	Dint	Duni	Ddes	Dreq	Total	
ID	4	4	12	7	3	30	
PD	0	0	0	0	0	0	%FE=100
RD	4	4	12	7	3	30	
RDRC	6.9	6.9	17	44	110	TRC >	804.8 cu

5. Comparison of Three Models

Quality assurance activity	Economy plan		Standard plan		Comprehensive plan	
	Percentage of removed defects	Cost of removing defects (cost units)	Percentage of removed defects	Cost of removing defects (cost units)	Percentage of removed defects	Cost of removing defects (cost units)
Requirements specification review	---	---	7.5%	7.5	9%	9
Design inspection	---	---	---	---	28.7%	35
Design review	25%	36.3	21.3%	21.3	7.4%	7.1
Code inspection	---	---	---	---	24.4%	31.8
Unit test - code	---	---	25.6%	50	4.2%	5.6
Integration test	---	---	17.8%	66.3	9.8%	18.8
Documentation review	---	---	13.9%	38.9	9.9%	10.9
System test	45%	48.9	7.0%	50.9	4%	13.2
Total for internal quality assurance activities	**70.0%**	**85.2cu**	**93.1%**	**234.9cu**	**97.4%**	**131.4cu**
Defects detected during operation	30.0%	804.8	6.9%	139.2	2.6%	22
Total	**100.0%**	**890cu**	**100.0%**	**374.1cu**	**100.0%**	**153.4cu**

6. Comparison Rules

- Quality Assurance(QA) activities should be performed **as early as possible**
- Additional QA activities prove **effective in reducing the percentage of defects**
- Additional SQA efforts, results in **improved defect removal effectiveness**
- Proves to reduce the percentage of “passed” defects, where **the costs of removing a detected defect are much lower** than in the next development phases